Cloud Computing Architecture

Dr. Subrajeet Mohapatra
Assistant Professor
Department of Computer Science & Engineering,
BIT Mesra, Ranchi
Jharkhand

Cloud Computing Architecture

Introduction

- Utility-oriented data centers are the first outcome of cloud computing, and they serve as the infra- structure through which the services are implemented and delivered.
- Any cloud service, whether virtual hardware, development platform, or application software, relies on a distributed infrastructure owned by the provider or rented from a third party.
- Cloud computing can be implemented using a datacenter, a collection of clusters, or a heterogeneous distributed system composed of desktop PCs, workstations, and servers.
- According to the specific service delivered to the end user, different layers can be stacked on top of the virtual infrastructure (Virtual machine manager, a development platform, or a specific application middleware).

Introduction

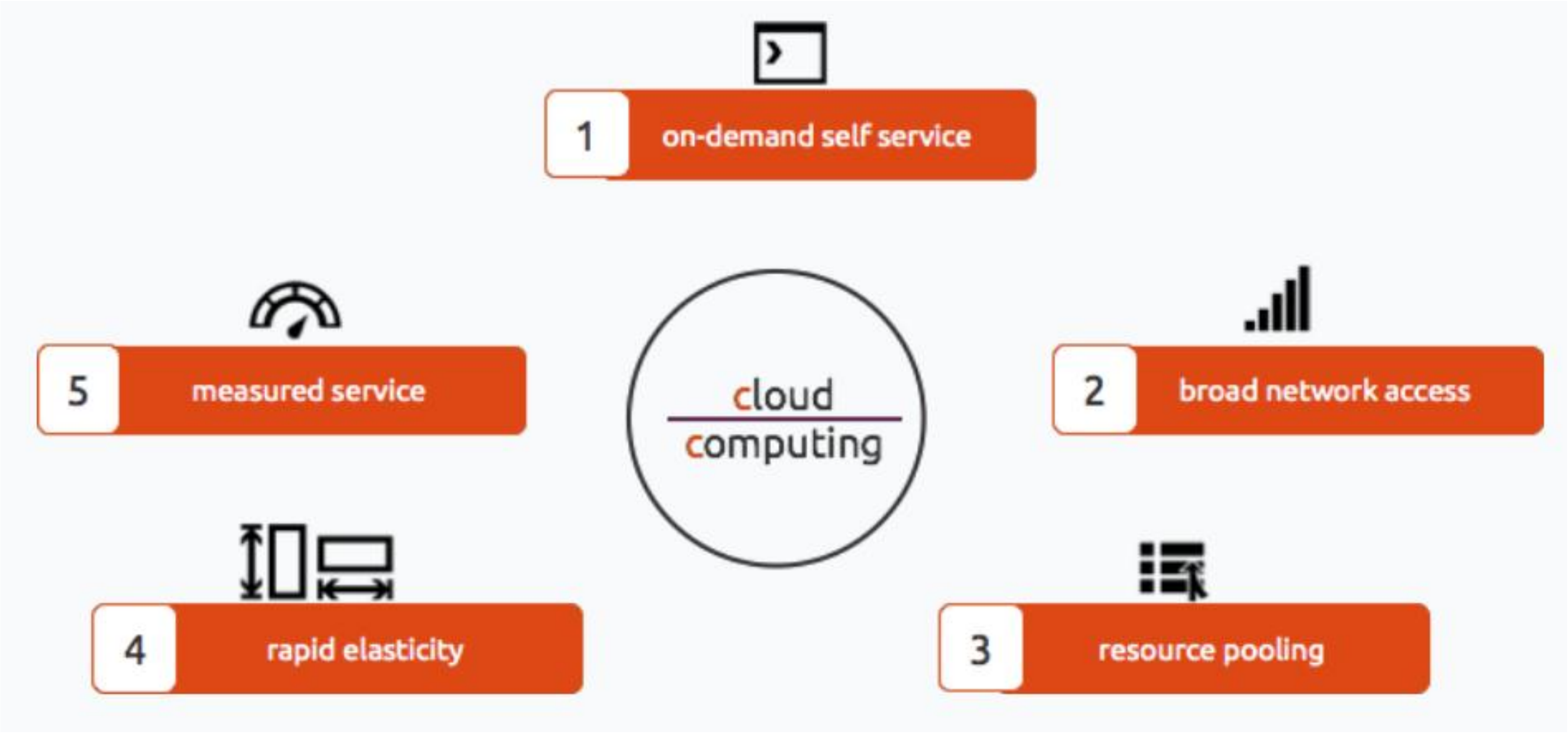
- The cloud computing paradigm emerged as a result of the convergence of various existing models, technologies, and concepts that changed the way we deliver and use IT services.

  Cloud computing is a utility-oriented and Internet-centric way of delivering IT services on demand.

  Cloud computing refers to storing and accessing data and programs on remote servers that are hosted on internet instead of computer's hard drive or local server.

- These services cover the entire computing stack: from the hardware infrastructure packaged as a set of VMs to S/W services such as development platforms and distributed applications.

Essential Characteristics



Al cloud computing based products, solutions and services must fulfill the above five criteria's.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Architecture

- Cloud computing architecture refers to the components and sub components required for cloud computing.
- These component typically refer to:
  - Front end (Thick client, Thin client)
  - Back end Platforms (Servers, Storage)
  - Cloud based delivery and a network (Internet, Intranet, Inter cloud).

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Thick Client vs. Thin Client

Thin Client :

- An application that runs on a server based computing environment.
- Works by connecting to a remote server based environment, where most applications and data is stored.
- Server performs most of the tasks like computations and calculations.
- More secure than thick client systems when it comes to security threats.
- System management is much easier as there are centralized servers.

Thick client

- A software that implements its own features and not dependent on server's applications.
- May connect to servers but remains mostly functional when disconnected.
- They have their own operating system and software applications.
- Have high flexibility and high server capacity.
- Have more security threats and are less secure than thin clients.

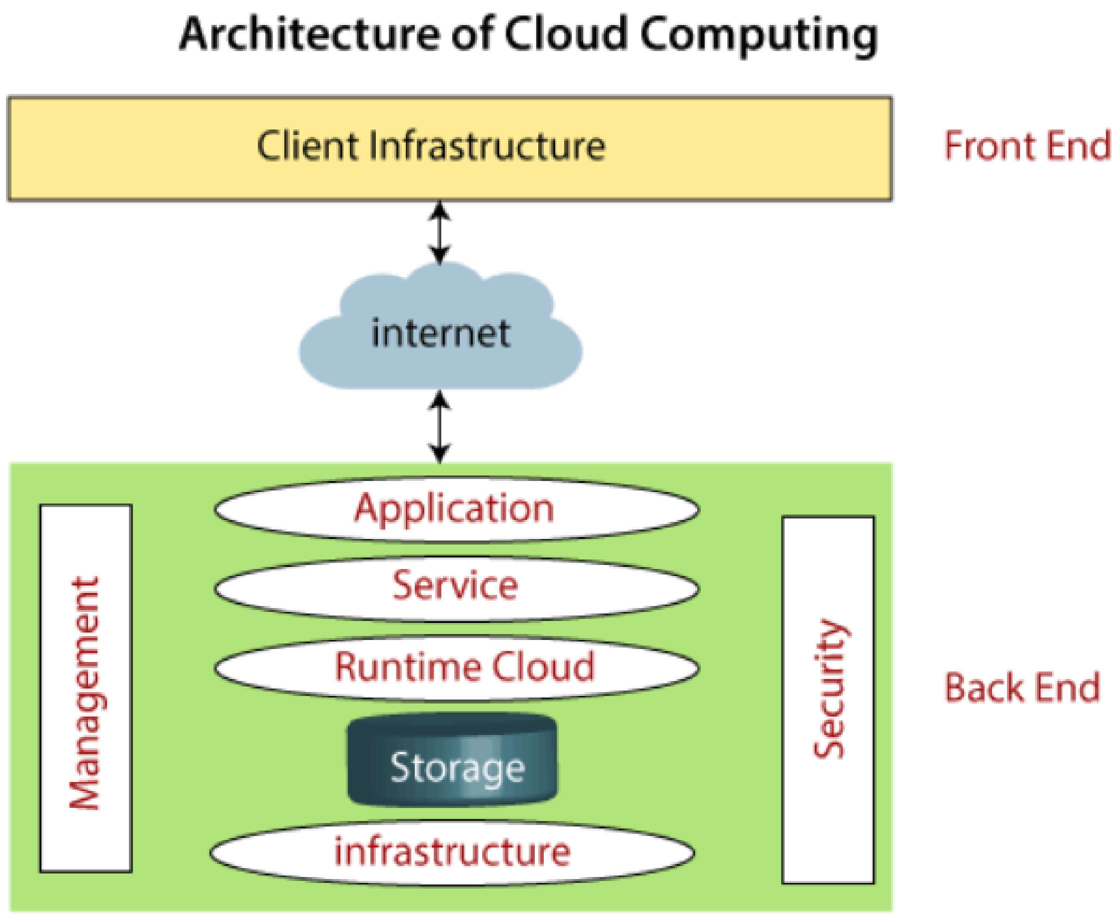
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Thick Client vs. Thin Client

Thin Client vs Thick Client		
	Thin Client	**Thick Client**
Definition	Software that relies on a remote server such as a cloud platform for its features.	Software that runs at least some features directly on your device.
Offline	Functions mostly don't work	Functions mostly work
Local Resources	Generally consumes few local resources such as disk, computing power and memory.	Generally consumes more local resources
Network Latency	Functionality may depend on a fast network connection.	Functionality may work without a connection or with a slow connection.
Data	Data is typically stored on servers.	Data may be stored locally.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Architecture (Simpler)



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Components of a Cloud Computing Architecture

Front End

- Used by the client (cloud user) and contains client-side interfaces and applications that are required to access the cloud computing platforms.
- Includes web servers (including Chrome, Firefox, internet explorer, etc.), thin & fat clients, tablets, and mobile devices.

Back End

- The back end is used by the service provider.
- It manages all the resources that are required to provide cloud computing services.
- Includes a huge amount of data storage, security mechanism, virtual machines, deploying models, servers, traffic control mechanisms, etc.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

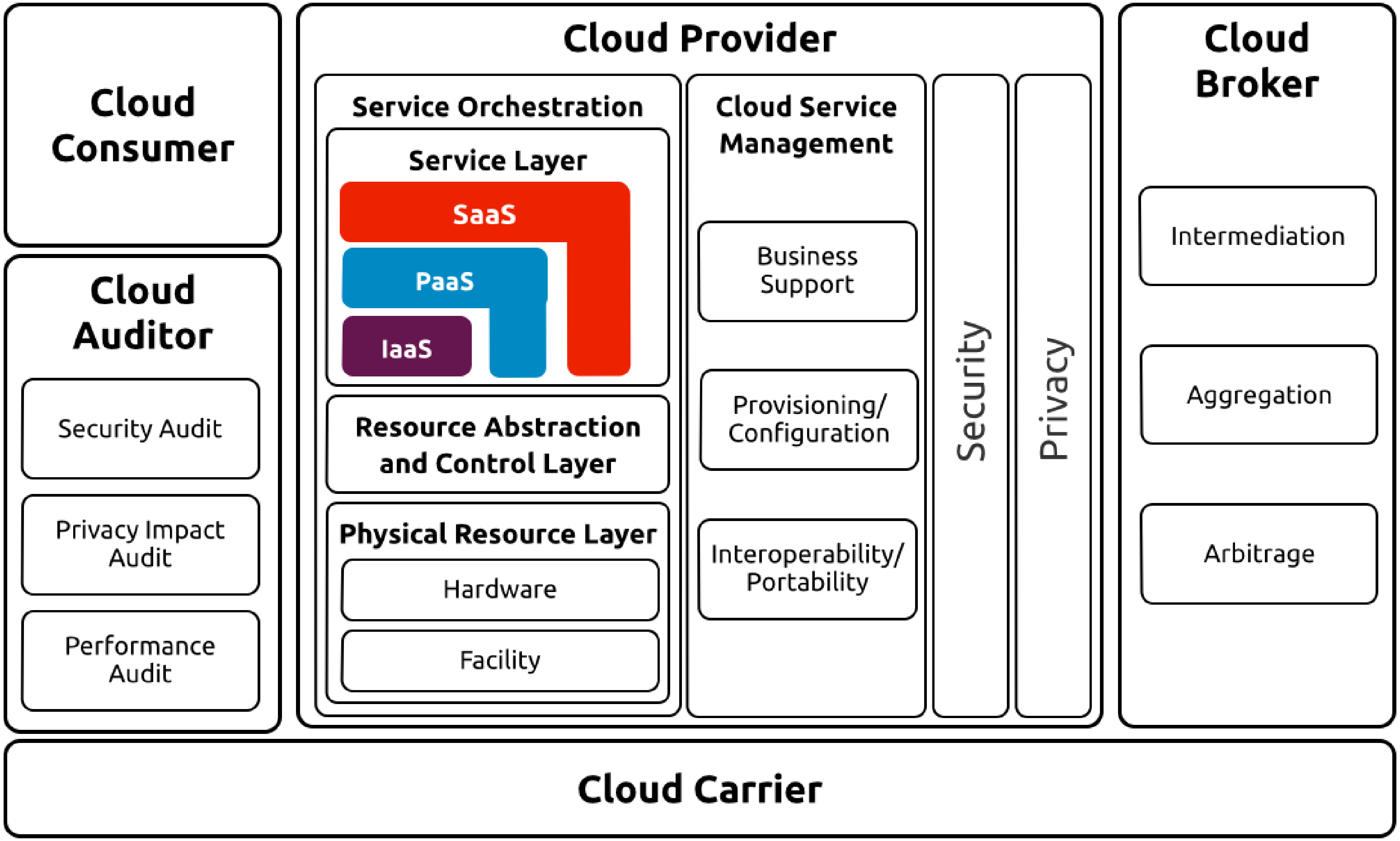
Components of a Cloud Computing Architecture

- **Client Infrastructure :** A front end component that provides GUI to interact/access cloud services.
- **Application :** Refers to any software or platform that a client wants to access.
- **Service :** Cloud computing offers three type of services: SaaS, PaaS, and Iaas.
- **Runtime Cloud :** Provides execution and runtime environment to the VM's.
- **Storage Infrastructure:** Provides storage capacity in the cloud to store and manage data.
- **Cloud Infrastructure :**
    - Provides services on the host level, application level, and network level.
    - Cloud infrastructure includes H/W and S/W components i.e. servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.

Components of a Cloud Computing Architecture

- **Management:** Includes managing components i.e. application, service, runtime cloud, storage, infrastructure, and other security issues in the backend and establish coordination between them.
- **Security :** An in-built back end component of cloud computing which implements security mechanism.
- **Internet :** An medium through which front end and back end can interact and communicate with each other.

Cloud Computing Reference Architecture (Detailed)



Cloud Computing Reference Architecture (Detailed)

- Reference architecture is a generic high-level conceptual model for discussing the requirements, structures, and operations of cloud computing.
- Current model is not tied to any specific vendor products, services, or reference implementation.
- Cloud Computing reference architecture defines five major actors : cloud consumer, cloud provider, cloud auditor, cloud broker and cloud carrier.
- Each actor is an entity (a person or an organization that participate in a transaction or process and/or performs tasks in cloud computing

Actors in Cloud Computing Reference Architecture

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, *Cloud Providers*.
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between Cloud Providers and Cloud Consumers.
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from Cloud Providers to Cloud Consumers.

Architectural Components of a Cloud Computing

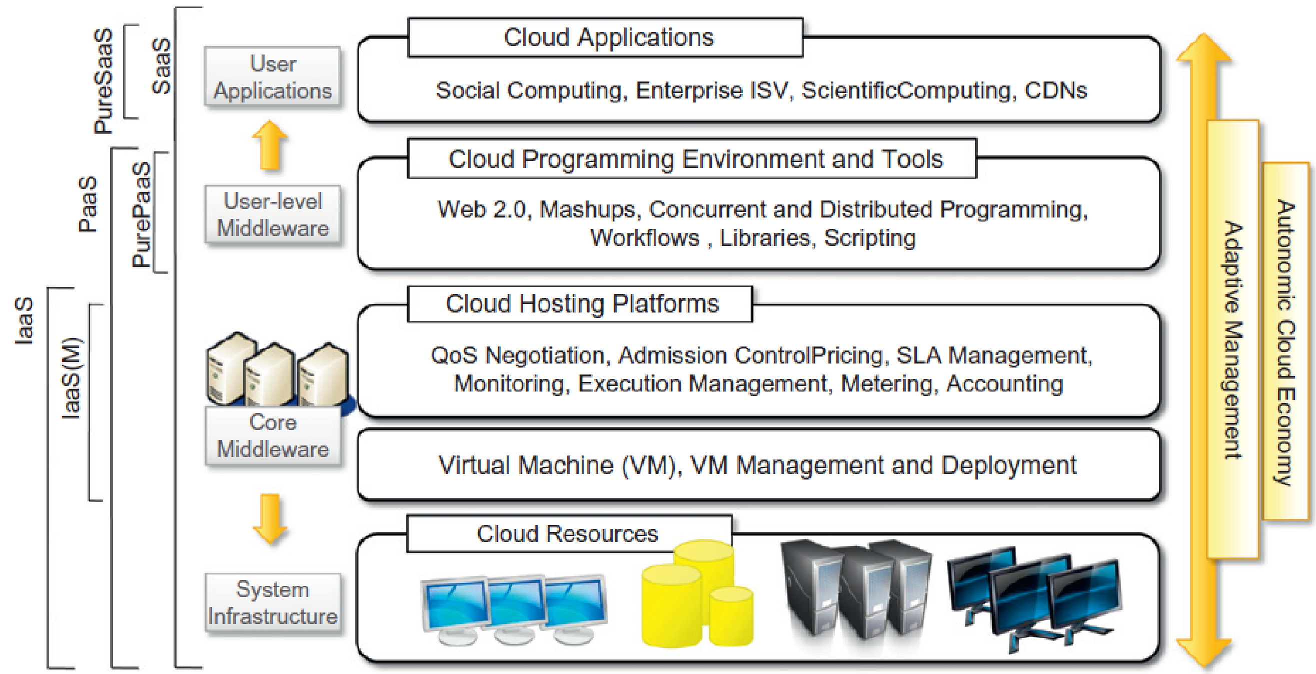
- **Service Deployment :**
  - Deployment model can be public, private, community and hybrid.
- **Service Orchestration :**
  - Refers to the composition of system components to support the cloud providers activities in arrangement, coordination and management of computing resources in order to provide cloud services to Cloud Consumers.
  - It consists of three layers i.e. service layer, resource abstraction and control layer and physical resource layer
- **Cloud Service Management :**
  - Includes all of the service-related functions that are necessary for the management and operation of those services required by cloud consumers.

Architectural Components of a Cloud Computing

- **Security :**
  - Cloud Computing systems need to address security requirements such as authentication, authorization, availability, confidentiality, identity management, integrity, audit, security monitoring, incident response, and security policy management.
  - Security in cloud computing spans across all layers of the reference model, ranging from physical security to application security.
- **Privacy**
  - Cloud providers should protect the assured, proper, and consistent collection, processing, communication, use and disposition of personal information and personally identifiable information in the cloud.
  - Its also the responsibility of *cloud auditor* to audit and report privacy in cloud computing architecture and actors involved.

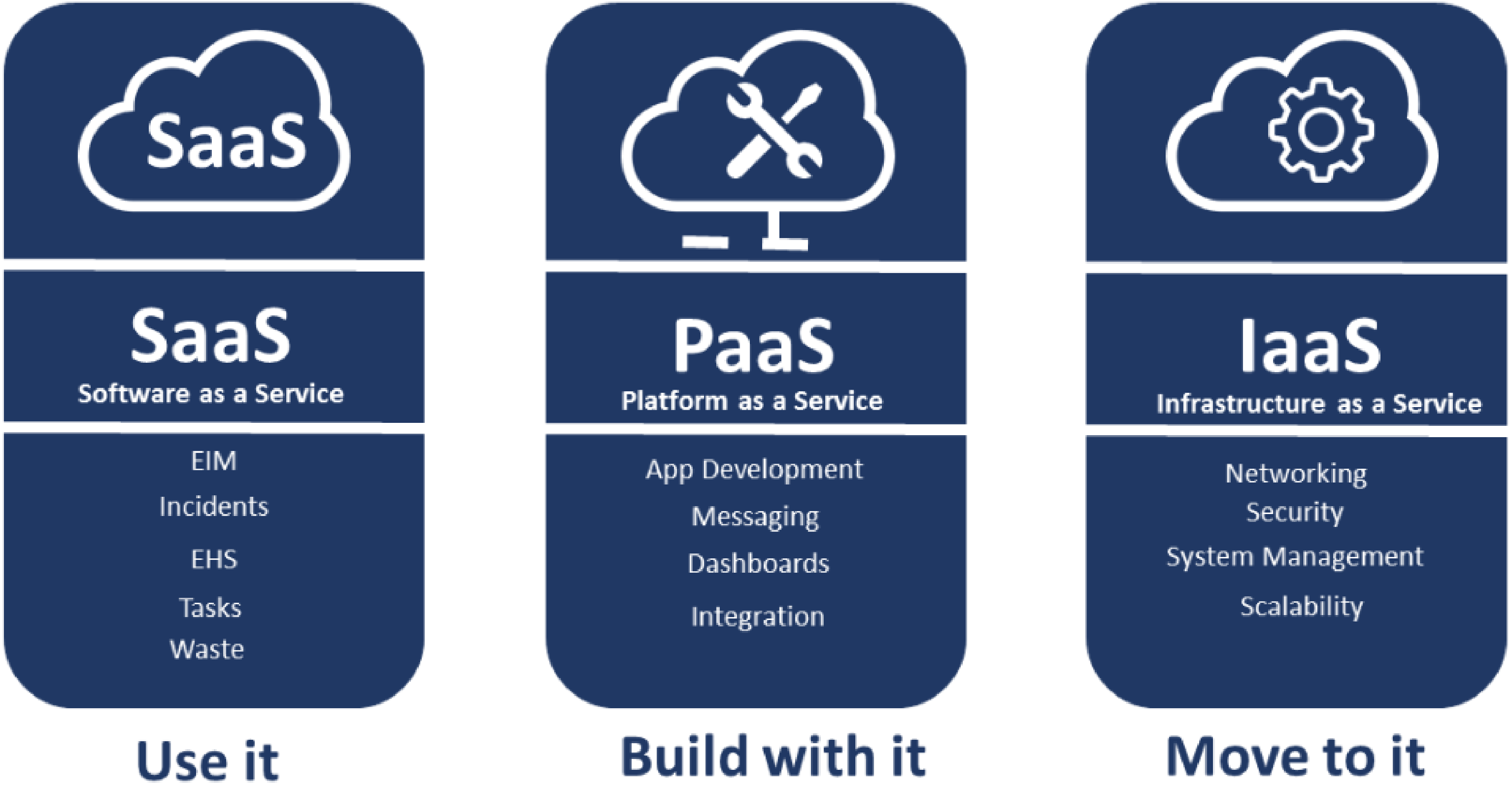
Cloud Computing Service Reference Model

- Cloud computing supports any IT service that can be consumed as a utility and delivered through a N/W, most likely the Internet.
- Such characterization includes quite different aspects: infrastructure, development platforms, application and services.



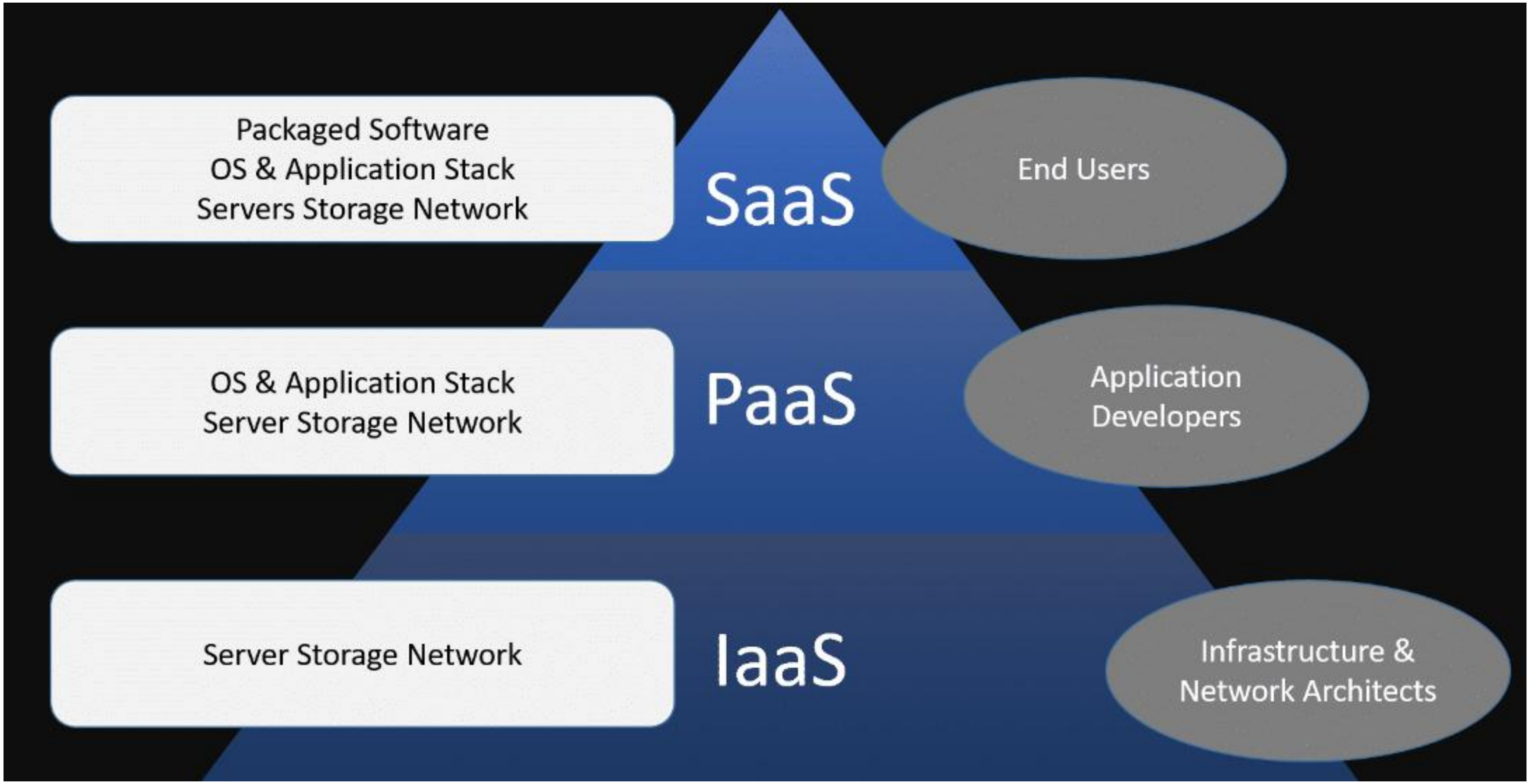
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Service Reference Model



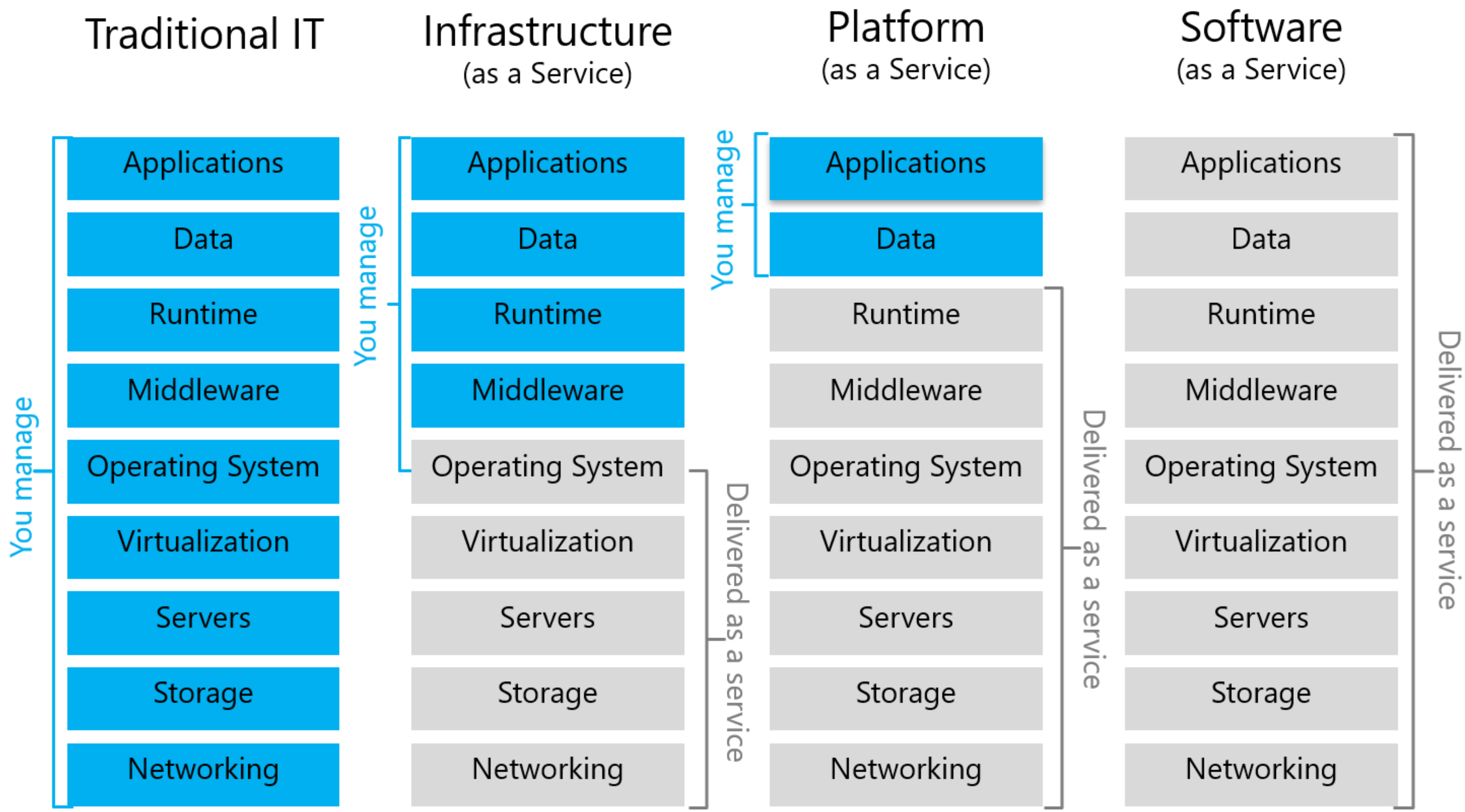
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Service Reference Model



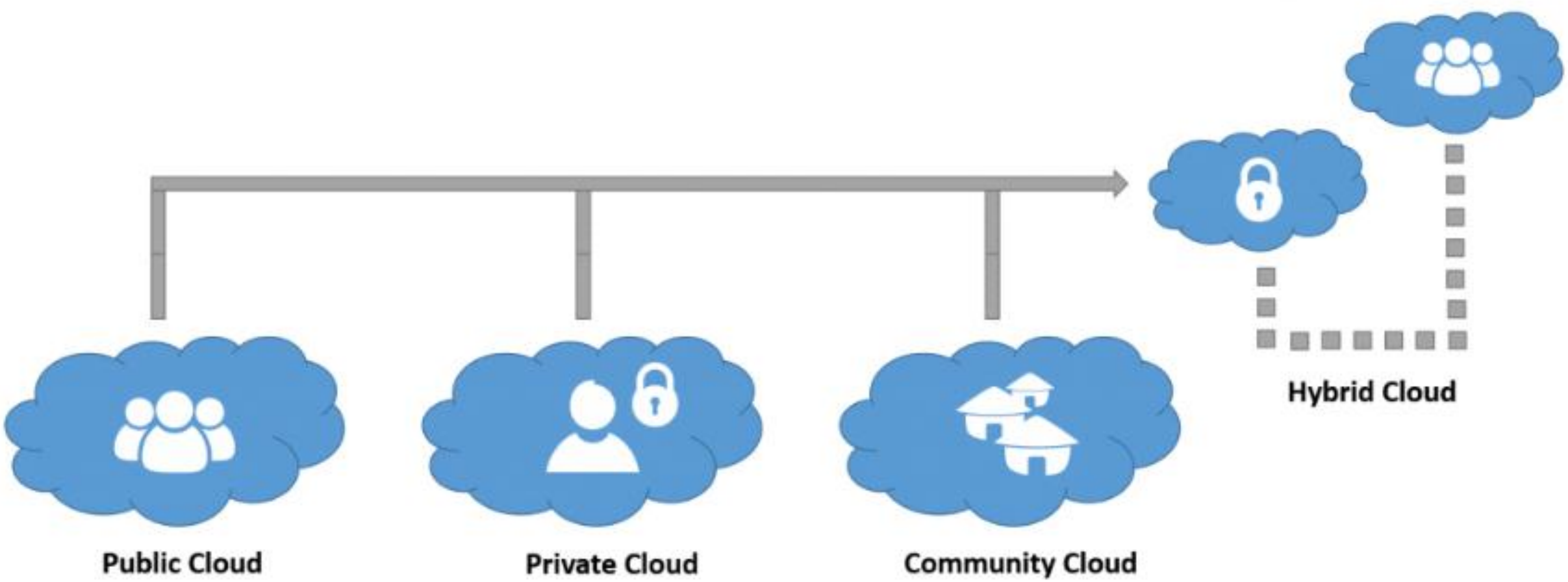
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Service Reference Model Comparison



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Deployment Models



Cloud Deployment Models

Characteristic	Public cloud storage	Private cloud storage	Hybrid cloud storage
Scalability	Very high	Limited	Very high
Security	Good, but depends on the security measures of the service provider	Most secure, as all storage is on-premise	Very secure; integration options add an additional layer of security
Performance	Low to medium	Very good	Good, as active content is cached on-premise
Reliability	Medium; depends on Internet connectivity and service provider availability	High, as all equipment is on premise	Medium to high, as cached content is kept on-premise, but also depends on connectivity and service provider availability
Cost	Very good; pay-as-you-go model and no need for on-premise storage infrastructure	Good, but requires on-premise resources, such as data center space, electricity and cooling	Improved, since it allows moving some storage resources to a pay-as-you-go model

Cloud Deployment Models Comparison

Deployment models	Holder	Security	Scalability	Cost
Private Cloud	Single private organization	Higher than other deployment models	Limited	High
Community Cloud	Two or more private organizations with identical requirements	Lower than Private Cloud and higher than Public and Hybrid Cloud	Limited	medium
Public Cloud	Cloud Service Provider (CSP)	Lower than other deployment models	Very High	Pay-per-use
Hybrid Cloud	CSP and private organizations	Lower than Private and Community Cloud and higher than Public Cloud	High	Pay-per-use

Cloud Based Services

Lecture 2, Module-4

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Based Services

- The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.
- Organizations offering these computing services are known as cloud service providers (CSP’s).
- CSP’s typically charge for cloud computing services based on amount of usage.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Stack

- Cloud computing has commonly been referred to as a “stack” because it typically encompasses many different types of services which have been built on top of each other under a “cloud”.
- Most cloud computing services fall into three broad categories :
  - Software as a service (Saas)
  - Platform as a service (PaaS)
  - Infrastructure as a service (IaaS)
  - Anything as a service (XaaS)
- Understanding cloud computing stack makes its easier to achieve the individual goals

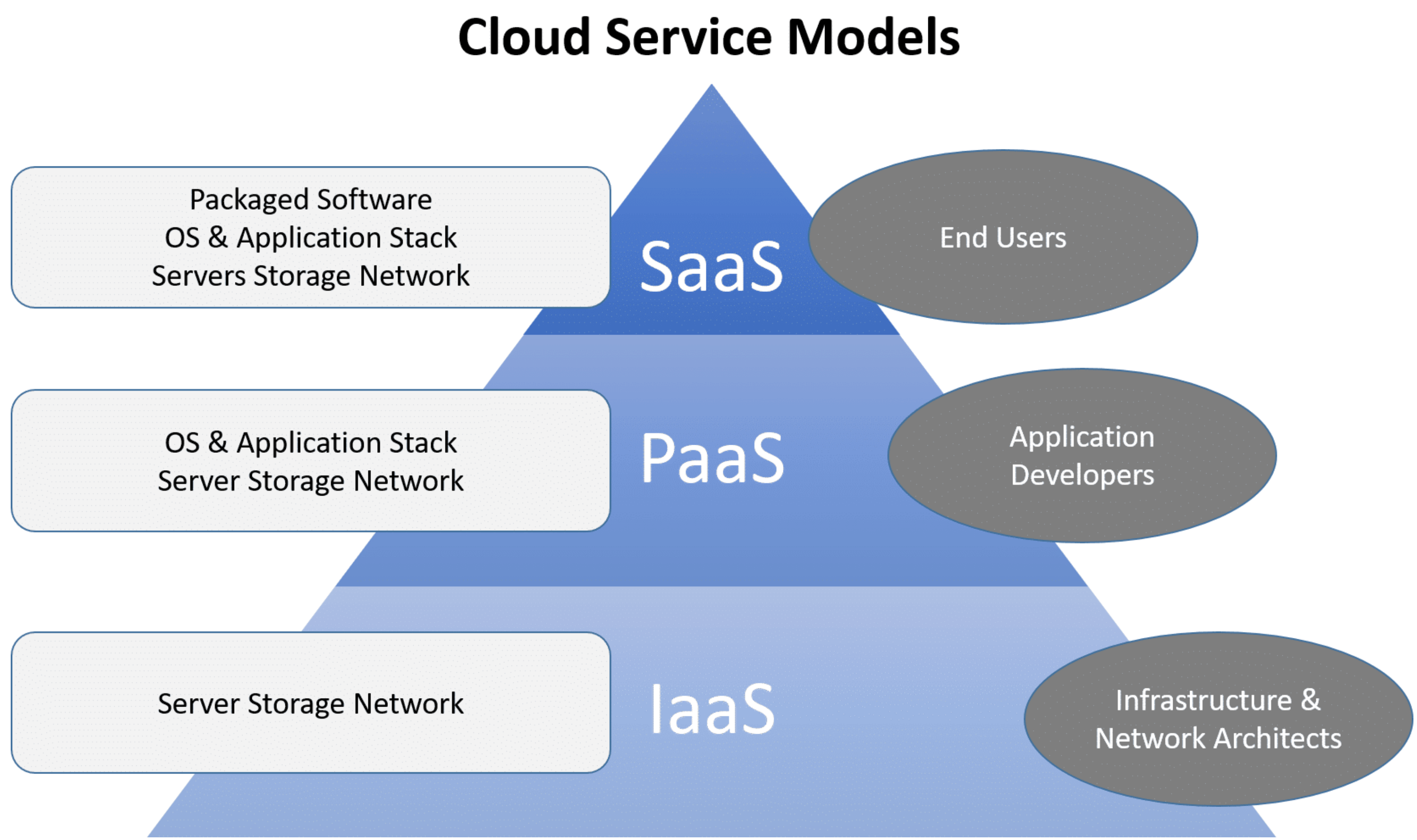
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Stack

Service Model	Main Access & Management Tool	Service content
SaaS Consume	Web Browser	Cloud Applications Social networks, Office suites, CRM, Video processing
PaaS Build On it	Cloud Development Environment	Cloud Platform Programming languages, Frameworks, PaaS Mashups editors, structured data
IaaS Migrate On it	Virtual Infrastructure Manager	Cloud Infrastructure Compute Servers, Data Storage, Firewall, Load Balancer

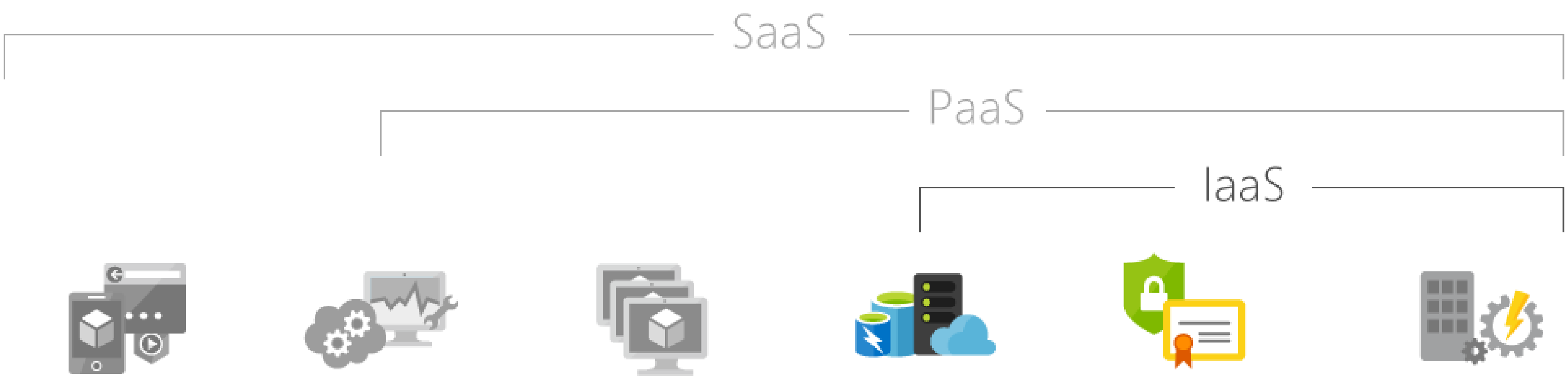
Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Stack



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Cloud Computing Stack (Horizontal View)



Software as a Service

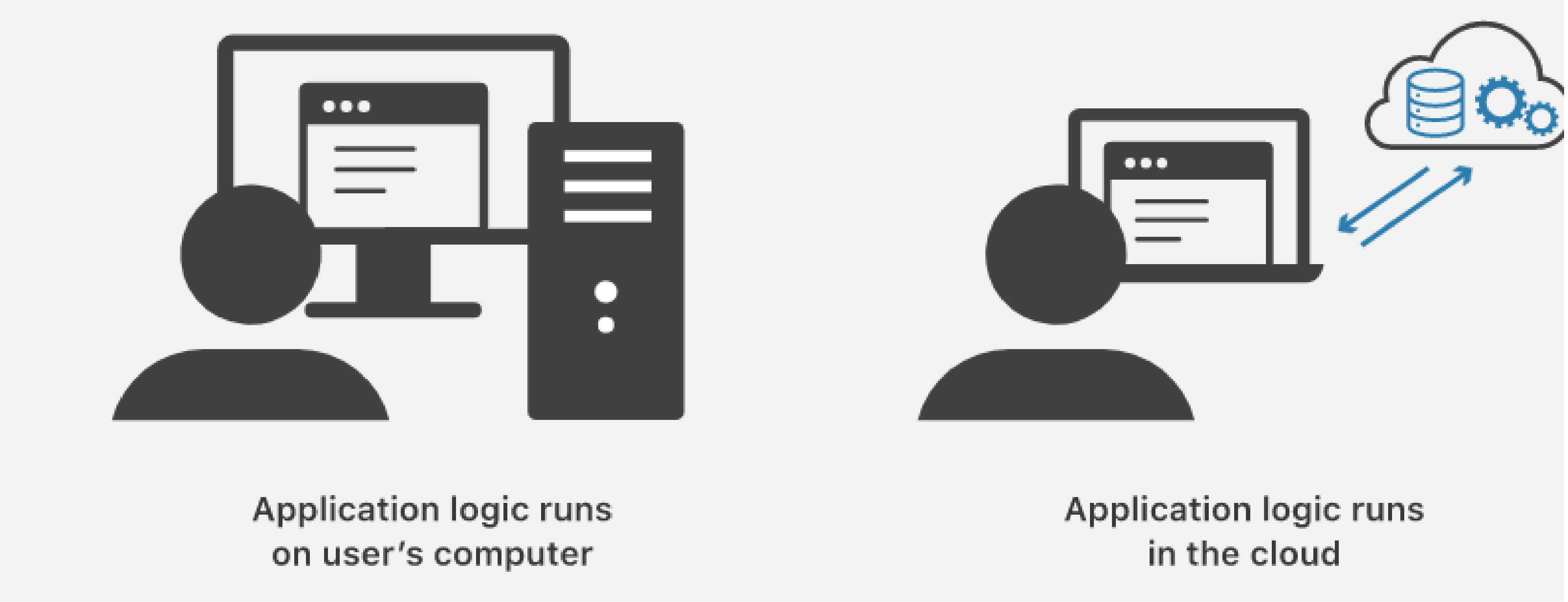
Introduction

- SaaS is a software delivery model wherein the software is delivered to the user over the internet in a subscription-based or pay-as-you-go model.
- Allows continual updates, provides the ability to access the software nearly anywhere, and gives the option for the customer to stop or start using the software service at will.
- In the SaaS model software is deployed as a hosted service and accessed over the internet, as opposed to “On premise”.
- Traditional software distribution model was based on the concept of software as a product (Purchased and installed on personal computers)
- SaaS users subscribe to an application rather than purchasing it once and installing it.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

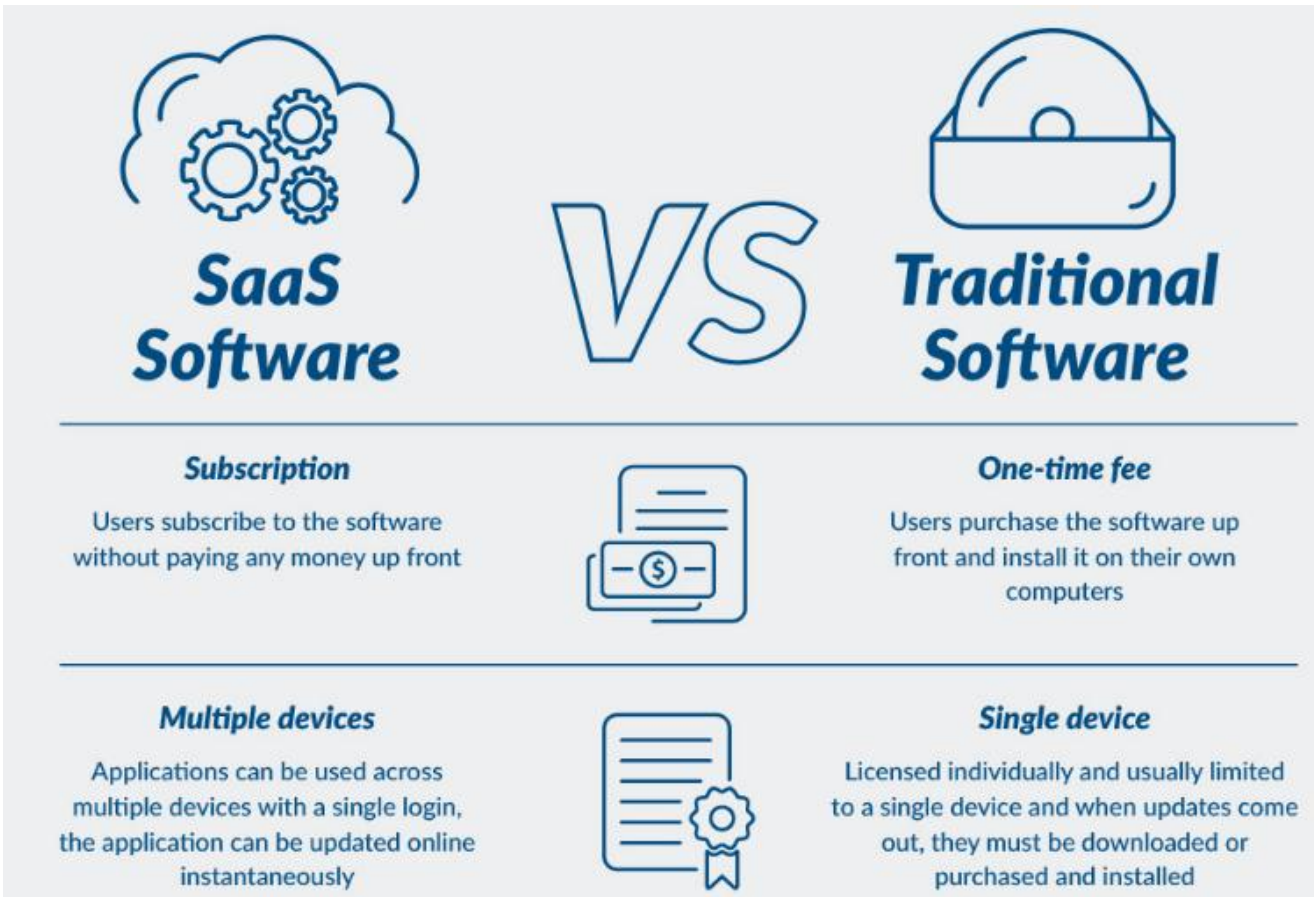
Installed Software Vs. SaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

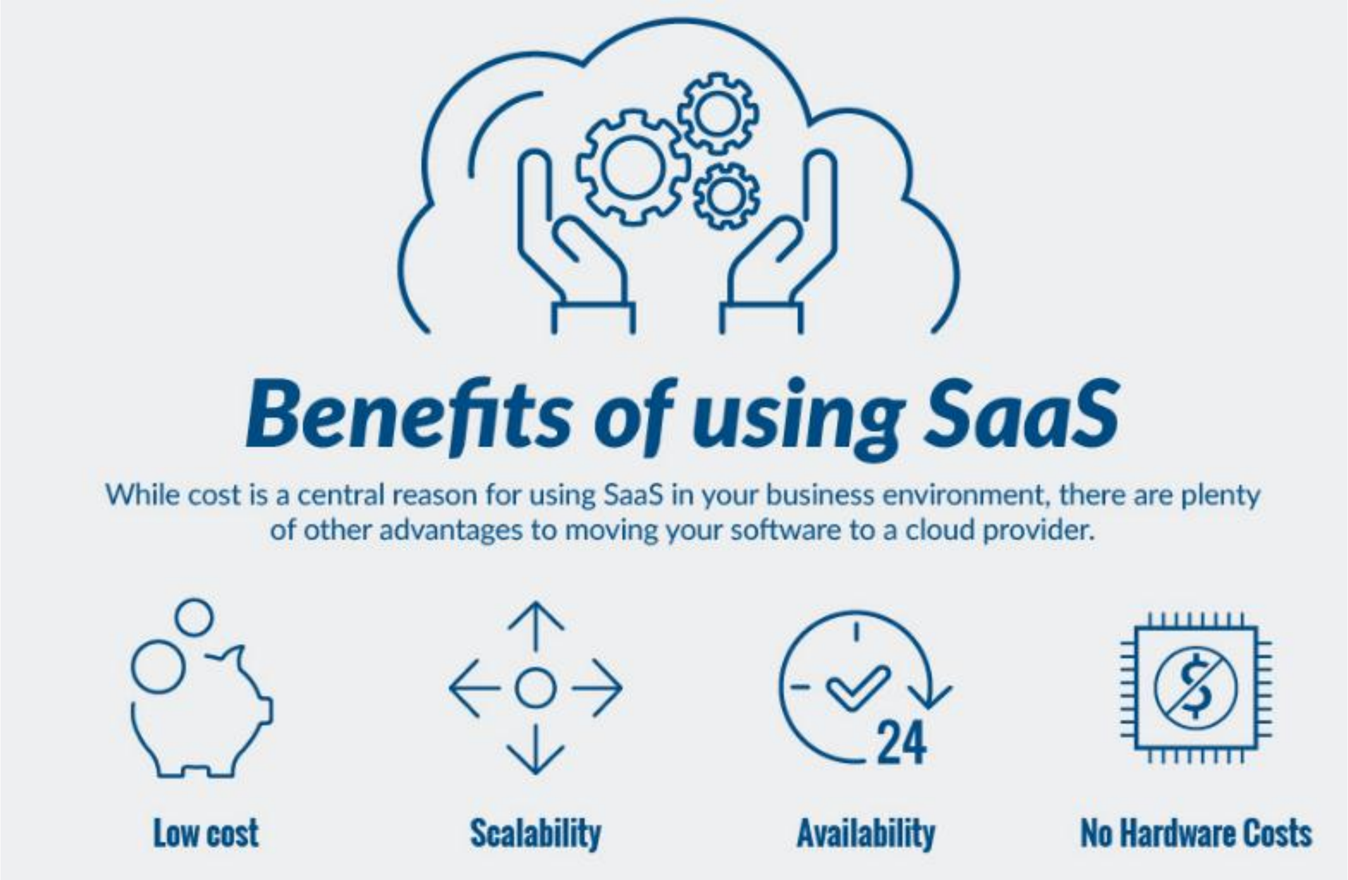
Installed Software Vs. SaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

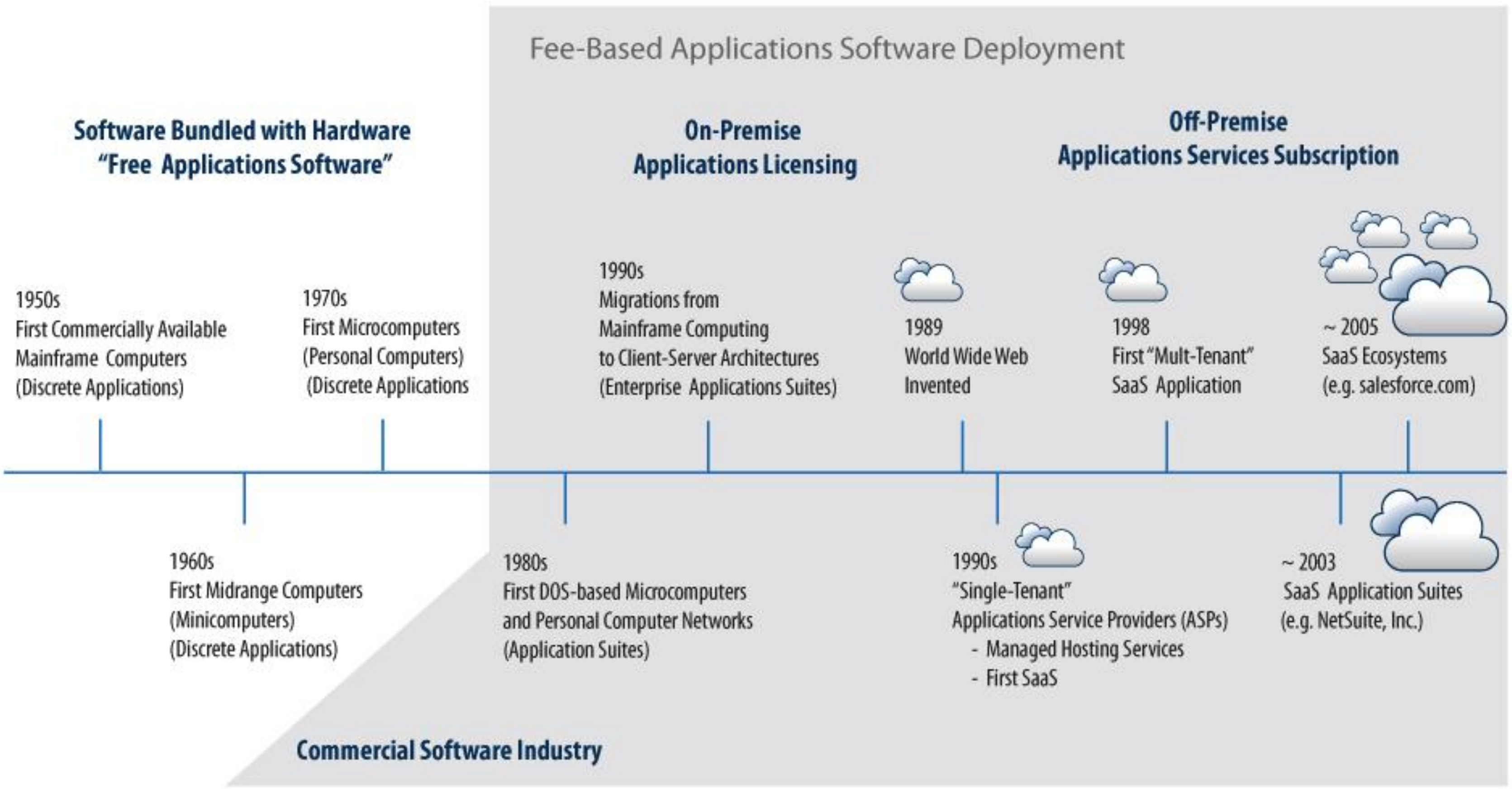
Benefits of SaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

History of SaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

Characteristics of SaaS

- Multi-tenancy model
- Automated provisioning
- Single Sign On
- Subscription based billing
- High availability
- Elastic Infrastructure
- Data Security
- Application Security
- Rate limiting/QoS
- Audit

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

Types of SaaS

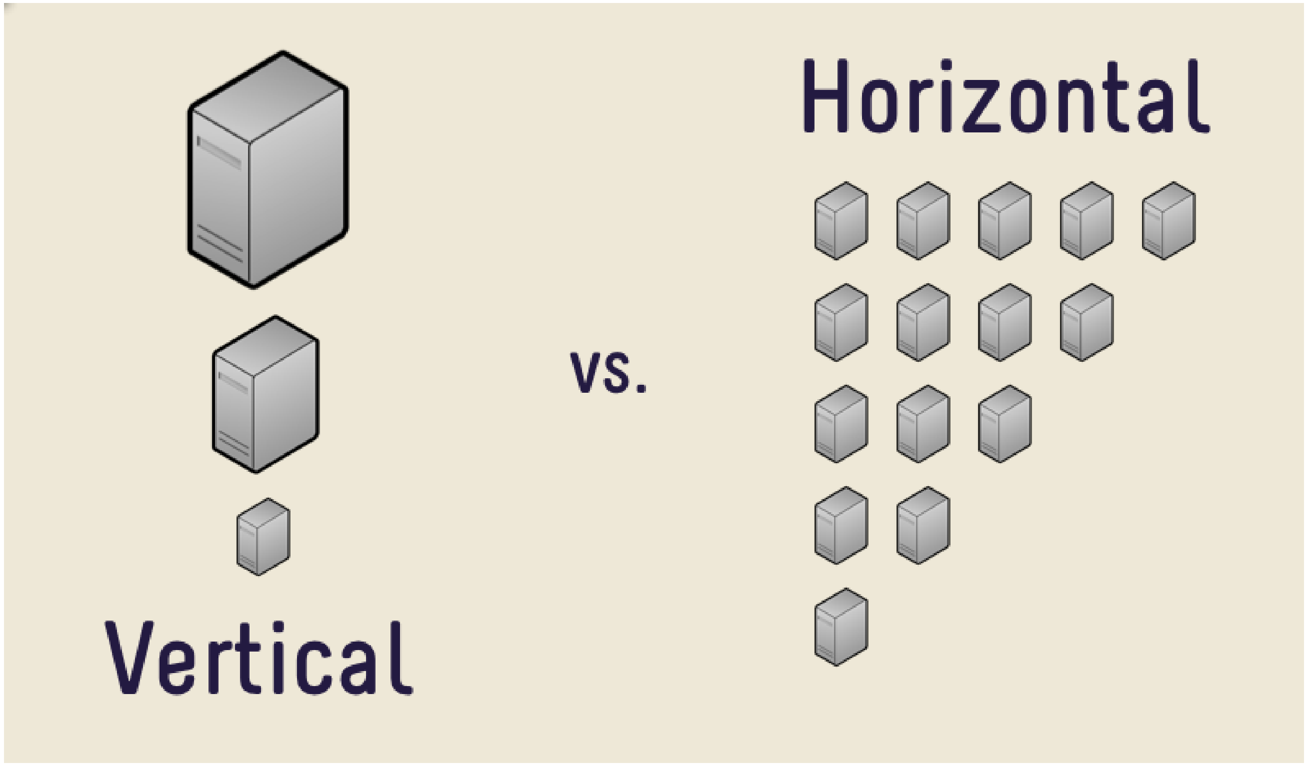
There are two main varieties of SaaS:

- **Vertical SaaS :** Software which answers the needs of a specific industry (e.g., software for the healthcare, agriculture, real estate, finance industries).
- **Horizontal SaaS :** The products which focus on a software category (marketing, sales, developer tools, HR) but are industry neutral.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

Types of SaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Software as a Service

Vertical SaaS vs. Horizontal SaaS

Horizontal SaaS	Vertical SaaS
Horizontal SaaS provides solutions based on the common needs of companies in different industries.	**Vertical SaaS** provides solutions to the exact problems of a specific industry.
Horizontal SaaS is not business-specific and can be used by various domains.	**Vertical SaaS** is business-specific and is designed for one particular domain.
Horizontal SaaS is easy to use and maintain.	**Vertical SaaS** is more complicated than horizontal software.
Horizontal SaaS caters to the needs of a broad category of audience like marketing, so customization options are limited.	**Vertical SaaS** supports extensive customization to satisfy the exact needs of the niche customers.
Horizontal SaaS takes time to adapt to industry-specific trends.	**Vertical SaaS** can quickly adapt to trends and serve the changing needs of the industry.
Horizontal SaaS relies on marketing strategies to reach a large audience.	**Vertical SaaS** relies on marketing strategies to reach a small audience and, since it's targeted, has a high chance of conversions.

Platform as a Service

What does Platform means?

- A platform is a group of technologies that are used as a base upon which other applications, processes or technologies are developed.

- In personal computing, a platform is the basic hardware (computer) and software (operating system) on which software applications can be run.

- Computing environment constitutes the basic foundation upon which any application or software is supported and/or developed.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Platform Standards

- The platform conforms to a set of standards that enable software developers to develop software applications for the platform.

- Standards allow owners and managers to purchase appropriate applications and hardware for running different applications.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service (Iaas to PaaS)

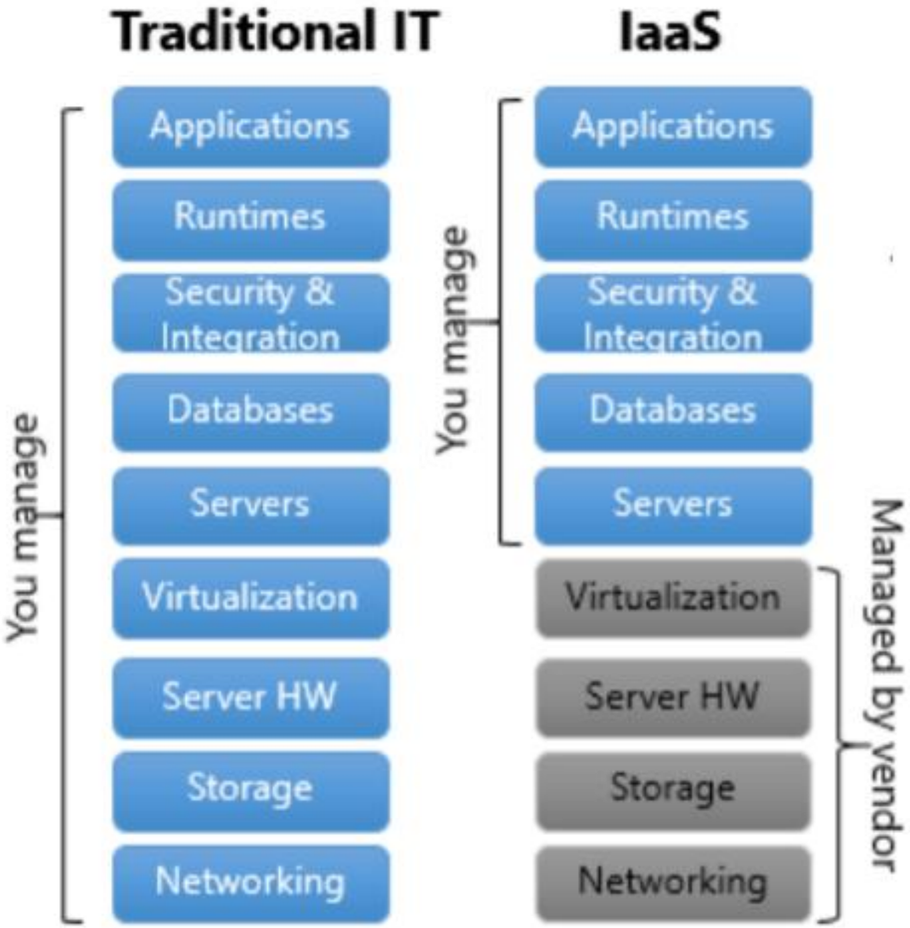
- IaaS provides VMs and resources such that IaaS vendors can segment resources for each user
- IaaS users are not required to invest in hardware
- IaaS provides features for optimum utilization of computing resources

Is it Enough !!

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

What Iaas can do?



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Iaas is Not Sufficient Enough

- IaaS provides many virtual or physical machines, but it cannot alter the quantity automatically
- Consumers might
  - Require to make automatic decisions from dispatching jobs to assign available resources
  - Need a running environment or a development and testing platform to design their applications or services

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

More Consumer Requirements

- Large-scale resource abstraction and management
- Requirement of large-scale resources on demand
- Running and hosting environment
- Automatic and autonomous mechanism
- Distribution and management of jobs
- Access control and authentication

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

We need more services

PaaS provides it for them

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

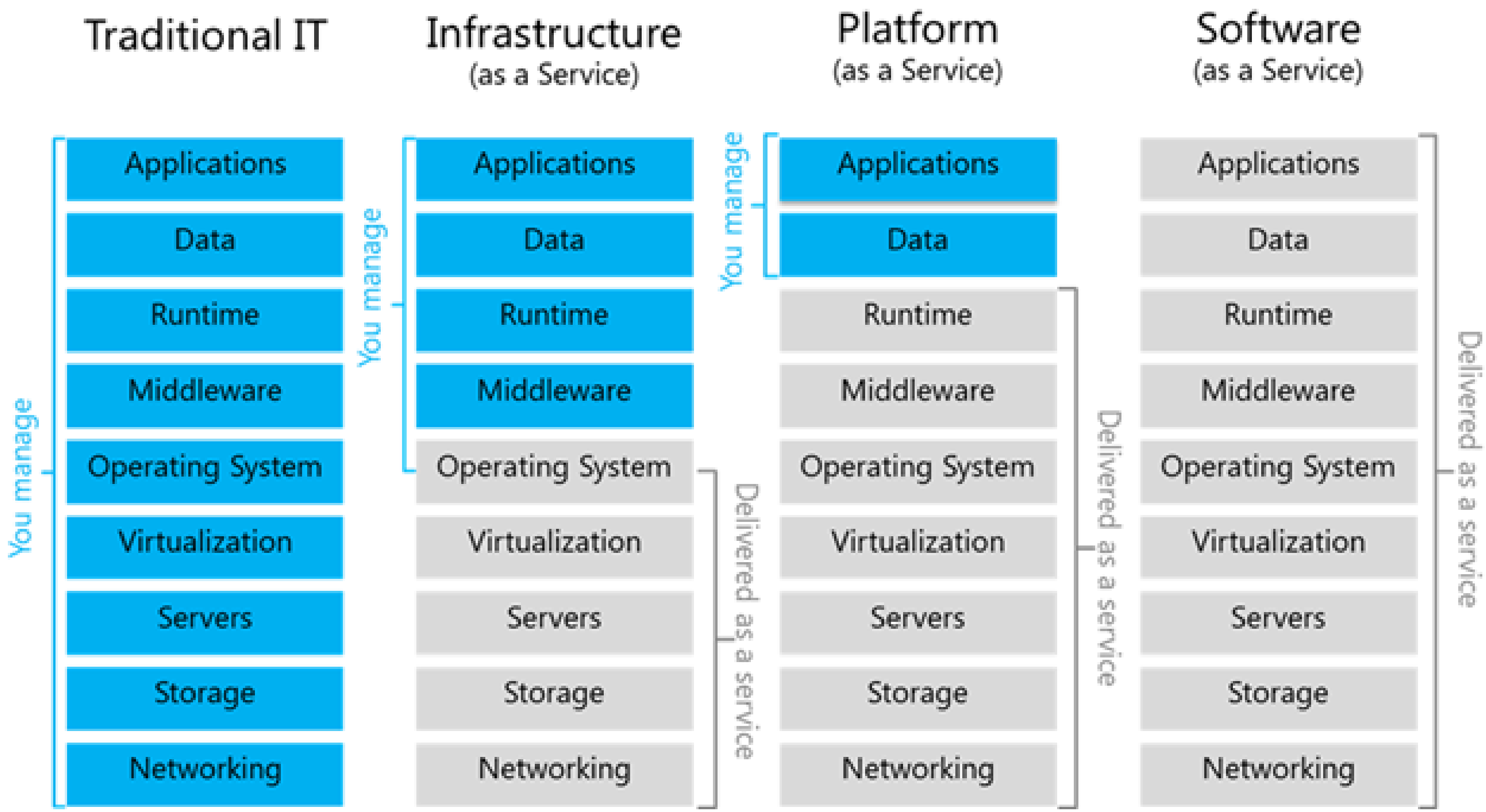
PaaS buys it for you

- PaaS provides a series of properties that can satisfy user's requirements
- Guarantees the quality of resources, services and applications

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

From IaaS to PaaS



Subrajeet Mohapatra
Department of CSE, BIT Mesra

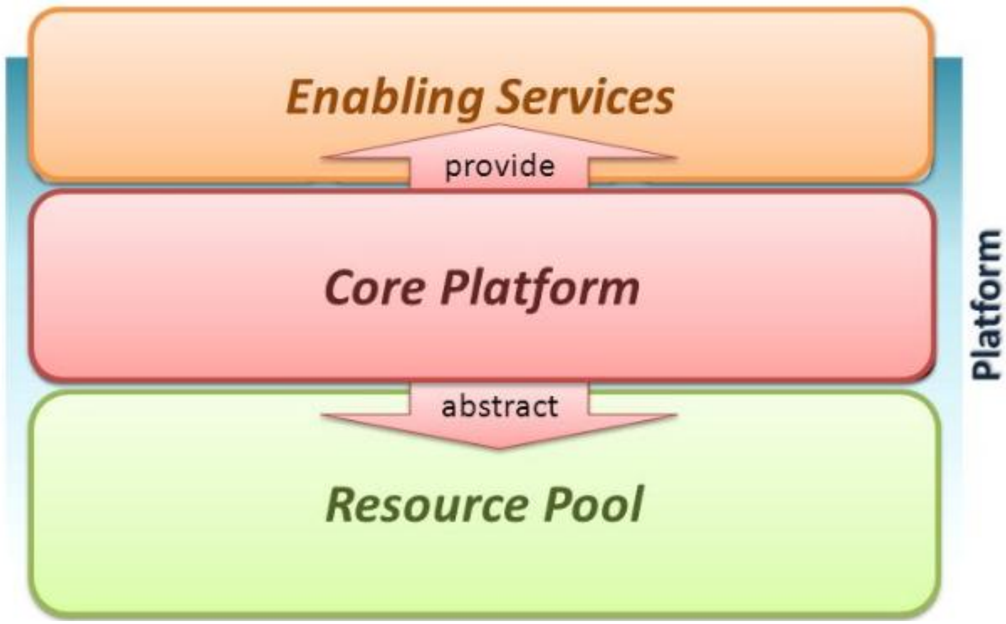
Platform as a Service

Platform as a service (PaaS) is a computing platform that abstracts the infrastructure, OS, and middleware to drive developer productivity.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

- Deliver the computing platform as a service
  - Developing applications using programming languages and tools supported by the PaaS provider
  - Deploying consumer-created applications onto the cloud infrastructure

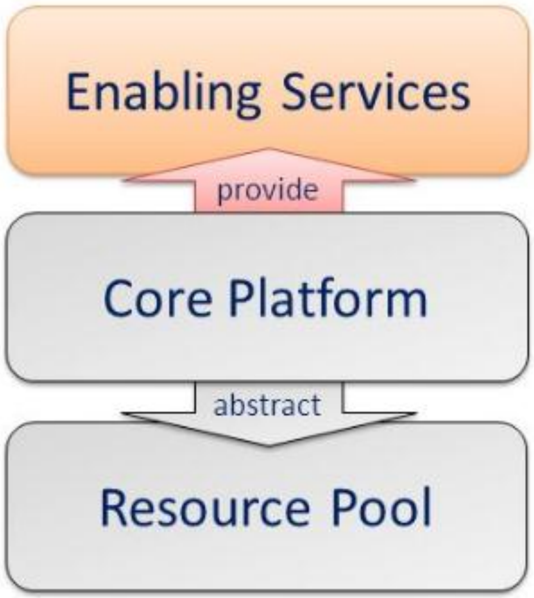


Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

- **Resource Pool :** The capacities to abstract and control all the underlying resources

  - Resource Pool dynamically provides an abstraction and consolidation of large-scale resources
  - Consumers can acquire and return resources from the resource pool on demand

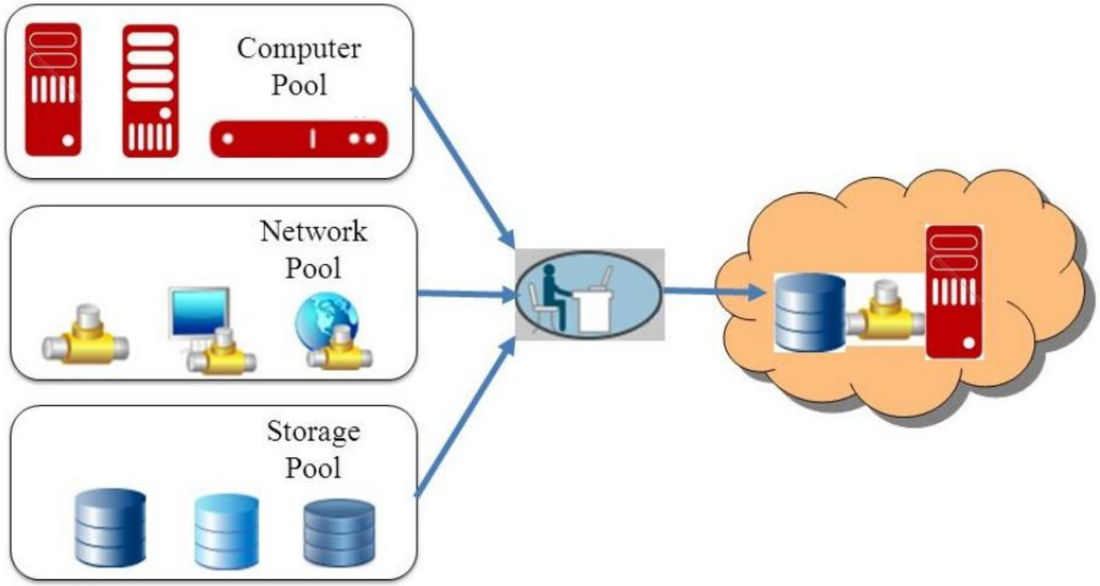


Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Resource Pool

- Reduce the complexity and responsibility of cloud infrastructure
- Provide the automatic management to provision resources
- Access resources from the resource pool on demand



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

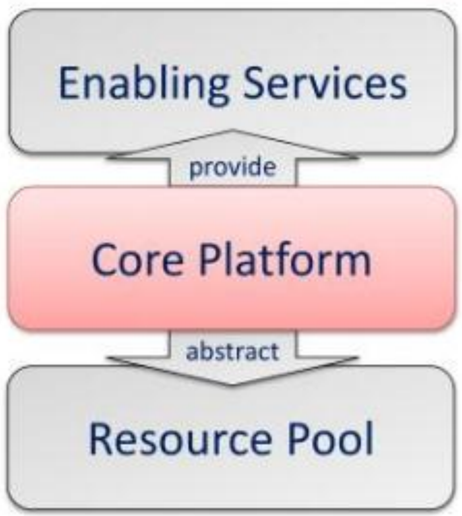
Resource Pool

- PaaS providers define the smallest unit of resource
    - 1GHz CPU computation ability
    - 1GB storage space
    - 1MB memory capacity …etc
- PaaS consumers can require units on their demand
- Consumers may not be aware of whether provided resource is dedicate or shared

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

- **Core Platform :** To provide a reliable environment for running applications and services (Enabling Services provide Core Platform abstract Resource Pool)
    - Core Platform provides basic functionalities of a PaaS environment
    - Act as a bridge between consumer and hardware



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Core Platform

- Reduce the responsibility of the runtime environment
- Based on the core platform developers develop their applications
- Developers are not required to take care about how to built, configure, manage and maintain the backend environment

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Core Platform

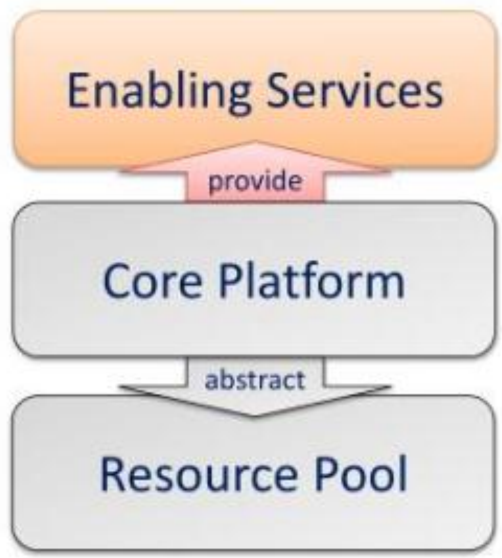
- PaaS providers can provide a runtime environment for the developer
- Runtime environment is an automatic control such that consumers can focus on their services
    - Dynamic provisioning
        - On-demand resource provisioning
    - Load balancing
        - Distribute workload evenly among resources
    - Fault tolerance
        - Continuously operating in the presence of failures
    - System monitoring
        - Monitor the system status and measure the usage of resources

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Enabling Services : To provide platform interfaces and services to drive the development productivities

- Enabling Services provide programming IDE and system control interfaces to access the PaaS environment
- Consumers can develop their applications through the APIs and development tools



Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Enabling Services

- Provide a development and testing platform for running developed applications on the runtime environment
- Reduce the responsibility of managing the development environment
- Decrease the development period

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Enabling Services

- Enabling Services are the main focus of consumers
- Consumers can make use of these sustaining services to develop their applications
  - Programming IDE
    - Integrate the full functionalities supported from the runtime environment
    - Provide some development tools, such as profiler, debugger and testing environment
  - System Control Interfaces
    - Make the decision according to some principles and requirements
    - Describe the flow of installation and configuration of resources

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Platform as a Service

Properties and Characteristics of PaaS

- Scalability
- Availability
- Manageability
- Performance
- Accessibility

Subrajeet Mohapatra
Department of CSE, BIT Mesra

PaaS Vendors

Windows® Azure™

heroku

Google™ App Engine

force.com™

Subrajeet Mohapatra
Department of CSE, BIT Mesra

PaaS Vendors

Summary

- PaaS is a magic box
- Request anything on demand, and return the rent of resources dynamically
- Automatically build an initial environment
- Support self-management with high quality of service and performance
- Provides fault tolerance and disaster recovery ability that make services more available and reliable
- Support the security property to limit malicious behavior in cloud environments
- Do not care about how PaaS works
- Pay as you go

Subrajeet Mohapatra
Department of CSE, BIT Mesra

Infrastructure as a Service

- Infrastructure as a service (IaaS) is a means of delivering computing infrastructure as an on-demand service.
- An instant computing infrastructure, provisioned and managed over the internet.
- It is one of the three fundamental cloud service models
- IaaS quickly scales up and down with demand, letting you pay only for what you use.
- It helps you avoid the expense and complexity of buying and managing your own physical servers and other data center infrastructure.

Subrajeet Mohapatra
Department of CSE, BIT Mesra

THANK YOU